Project Proposal - F1 Race Winner Prediction

What?

Formula 1 races are determined by a complex interaction of factors such as track type, weather, tire strategy, driver performance, and team competitiveness. While teams rely heavily on internal analytics, there is no simple tool for students, fans, or independent analysts that predicts race outcomes using publicly available data.

This project aims to build a machine learning model that predicts the **winning probabilities** for each driver before a Grand Prix. The model will use open data from the **F1DB repository** (races, results, qualifying, standings, etc.) to identify which variables—such as qualifying position, historical performance, team strength, and track characteristics—most strongly influence race outcomes.

The model's output will be a list of the **top 3 drivers with the highest chance of winning**, along with probability scores. The goal is not to replace expert judgment, but to **augment understanding** by transforming raw motorsport data into clear, data-driven insights.

Why?

My motivation for this project comes from a very personal shift in interest.
For a long time, I did not consider watching Formula 1 at all—it looked to me like "just fast cars driving in circles." That changed when I accidentally watched a race with friends. I realised very quickly that F1 is not only about speed; it is about **strategy, timing, tire choices, weather, track characteristics, and constant decision-making under pressure**. The depth and complexity of the tactics really triggered me.

At the same time, I am studying **AI and data analysis**, and I was actively looking for a project where I could apply those skills to a domain I genuinely care about. F1 felt like the perfect combination:

- it is rich in structured data,
- it is highly strategic,
- and it changes race by race, which makes prediction both challenging and interesting.

Another reason for choosing this topic is its **relevance for my future**. I am interested in a company that collaborates with Dutch F1, and I was already looking for a project that I could potentially use as part of my **internship candidacy**. Being able to show a concrete project that combines F1 and AI is a strong way to demonstrate both my passion and my technical skills.

From there, the idea evolved naturally:

If F1 is so dependent on many measurable factors, can I use AI to predict who is most likely to win the next Grand Prix?

This project is my way to connect three things that matter to me:

- My **new passion for F1**,
- My **AI and data analysis studies**,
- My **future career ambitions** in data-driven, sports-related or tech environments.

Who?

Project owner:

- Responsible for data collection, cleaning, EDA, model design, implementation, and reporting.

Stakeholder:

- A software developer with strong F1 knowledge, who acts as my domain expert and sparring partner.
- Helps validate whether the chosen variables and assumptions make sense in real racing contexts.
- Provides feedback on the practicality and realism of the model's predictions.

Teachers / Supervisors:

- My teachers will provide feedback on the technical quality, research approach, and iterations of the project.
- They will support me in refining the model, interpreting results, and improving the overall structure of the project.
- The combination of my own interest, a domain-knowledgeable stakeholder, and academic guidance ensures the project is grounded in both real F1 context and sound AI methodology.

When?

I have **7 weeks in total**. The process is based on a **data-first approach**: I started with data finding, scraping, and cleaning before finalizing the exact idea.

- **Weeks 1–3 (Completed): Data-First Phase**
  - Found and downloaded data from F1DB.
  - Cleaned and merged the main datasets.
  - Created the Domain Understanding.
  - Wrote this Project Proposal.
  - Started Exploratory Data Analysis (EDA) and initial visualizations for DAIA.
- **Week 4: Finish EDA + Define Features**
  - Complete EDA and visualizations.
  - Decide which features (variables) to include in the first version of the model.
  - Document insights about which factors seem most relevant.
- **Week 5: First Iteration – Baseline Model**
  - Implement a simple baseline prediction model (e.g., basic classification/regression based on grid position and standings).

    - Evaluate whether the model can predict winners better than random or naive baselines.
- **Week 6: Second Iteration – Add Complexity**
    - Extend the model with more factors (e.g., track type, historical performance, team form).
    - Improve the model's performance and evaluate again.
    - Discuss results with teachers and stakeholder, adjust assumptions if needed.
- **Week 7: Third Iteration + Finalization**
    - If possible, add or refine another factor (e.g., basic tire/strategy info or weather proxy).
    - Final evaluation and comparison across iterations.
    - Finalize visualizations, write conclusions, limitations, and suggestions for future work.

How?

Because the project is complex and F1 has many interacting variables, I will work in **iterations** instead of trying to model everything at once. The approach is:

Data-First & Preparation

- Start from the available data (F1DB).
- Clean, merge and understand what is realistically usable.
- Let the data guide what is possible.

Iterative Model Building

- **Iteration 1:** Simple model using only a few strong predictors (e.g., grid position, driver and team standings).
- **Iteration 2:** Add more context (track characteristics, historical performance at that circuit).
- **Iteration 3:** Where feasible, include strategy-related or proxy variables (e.g., pit stop info, weather if available).

Each iteration will include:

- training the model,
- evaluating results,
- getting feedback from teachers and the stakeholder,
- and then adjusting.

Exploration & Explanation

- Use EDA and visualizations to understand which variables appear most important.
- Analyze feature importance (where possible) to see *why* the model makes certain predictions.

Reporting & Reflection

- Summarize the final model performance and the differences between iterations.

- Reflect on what worked, what did not, and how the model could be improved in the future (e.g., integrating more detailed tire/strategy data or live inputs).

This iterative, step-by-step approach matches the complexity of the problem and allows me to build up the model gradually, with continuous feedback from both teachers and my stakeholder.